

Curriculum Vitae

Aarne Talman

PERSONAL DETAILS AND DATE OF CV

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Date of CV 30 September 2026

DEGREES

19.03.2024 Doctor of Philosophy (PhD; Finnish: filosofian tohtori, FT), third-cycle degree, Language Technology, University of Helsinki, Finland. Dissertation passed with distinction. Dissertation: *Towards Natural Language Understanding: Developing and Assessing Approaches and Benchmarks*. Supervisors: Prof. Jörg Tiedemann, Prof. Stergios Chatzikyriakidis (University of Crete) and Dr. Anssi Yli-Jyrä. Opponent: Prof. Vered Shwartz (University of British Columbia).
Awarding organisation: Faculty of Arts, University of Helsinki, P.O. Box 3 (Fabi-aninkatu 33), FI-00014 University of Helsinki, Finland.

01.01.2007 MSc in Computational Linguistics and Formal Grammar, with Distinction (second-cycle degree), King's College London, United Kingdom. Dissertation: *Path Grammars and the Generative Capacity of Dynamic Syntax*.

31.06.2005 BSc in Philosophy, First Class Honours (first-cycle degree), London School of Economics and Political Science, United Kingdom. Thesis: *Gödel's Incompleteness Theorems and the Limitations of Artificial Intelligence*.

CURRENT EMPLOYMENT

10/2025– Senior Manager, Software Development, AMD Silo AI, Helsinki, Finland (full-time). Responsible for AI software development, with a focus on large-model inference infrastructure and performance on AMD GPUs.

Career stage II (recognised researcher, R2) in the four-stage research career model, adapted: research carried out alongside industry employment.

Secondary positions

2024– Hourly-paid teacher (tuntiopettaja), Department of Digital Humanities, University of Helsinki.

2023– Visiting Scholar, Department of Digital Humanities, University of Helsinki. Research on natural language understanding and the evaluation of language models.

PREVIOUS WORK EXPERIENCE

2024–2025 Senior Manager & EMEA ML Lead, Accenture, Helsinki. Led the Machine Learning and Computational Science practice and the Model Customization capability, EMEA Centre for Advanced AI.

2023–2024 Head of R&D, SiloGen (Silo AI), Helsinki. Led research and development of open large language models.

2023 Senior Manager, Accenture, Helsinki. Technology strategy advisory on AI and cloud computing.

2021–2023 Senior AI Engineer (2021–2022), Lead AI Engineer (2022) and Lead AI Scientist (2023), Silo AI, London and Helsinki. Natural language processing, search, speech recognition and natural language understanding.

2020–2021	UK CTO and Global ML Practice Lead, Nordcloud, United Kingdom.
2019–2020	Founder and CEO, Basement AI, Finland and United Kingdom; research lab and consultancy in NLP and machine learning.
2018–2024	Doctoral candidate, Language Technology, University of Helsinki. Member of the ERC-funded project <i>Found in Translation: Natural Language Understanding with Cross-lingual Grounding</i> (FoTran, 2018–2023; PI Prof. Jörg Tiedemann).
2012–2018	Management consulting: Associate Director (2016–2018) and Senior Consultant (2015–2016), Gartner; Consultant (2012–2015), Accenture; Helsinki.
2011–2012	Research Student, London School of Economics, funded by the Alfred Kordelin Foundation. Reliability of non-linear models in economics and climate science.
2006–2011	Software and R&D management: Manager, Architecture and System Design (2008–2011) and Product Manager, Search (2009–2011), Nokia; Systems Analyst, Tieto (2006–2008); Software Developer, Valuatium (2006).

RESEARCH FUNDING AND GRANTS

2023	European Commission, DeployAI consortium (28 MEUR in total): work package lead, 1.15 MEUR.
2023–2024	Business Finland R&D funding, 6 MEUR, <i>Methods and Tools for Trustworthy and Reliable Large Language Models</i> : responsible leader.
2023	LUMI supercomputer, Extreme Scale Access, 12 million GPU hours (peer-reviewed) for large language model pre-training: co-PI.
2019	Travel grant, Lisbon Machine Learning School (LxMLS).
2011–2012	Alfred Kordelin Foundation, one-year personal research grant (London School of Economics).
2005–2006	Arts and Humanities Research Council (UK), Research Preparation Masters Scheme: full studentship and maintenance grant.

RESEARCH OUTPUT

Open-source software and data, released under open licences:

2024	Contributor to Poro 34B, an openly released multilingual large language model for Finnish, English and code.
2023	Uncertainty-aware NLI with stochastic weight averaging (code).
2021–2022	NLI Data Sanity Check and NLU Dataset Diagnostics: code and corrupted NLI and GLUE datasets for testing whether models rely on the intended information.
2019	Prosody: system for predicting prosodic prominence from text (MIT licence), and the Helsinki Prosody Corpus, prosodic annotations for LibriTTS (CC BY 4.0).
2018	HBMP: natural language inference system with the HBMP sentence encoder (MIT licence).

RESEARCH SUPERVISION AND LEADERSHIP EXPERIENCE

Supervision	Master’s theses, Language Technology, University of Helsinki: 1 completed and 1 in preparation. 2025: Jaakko Kuurne, <i>Fine-tuning Open-weight Large Language Model for Expense Classification</i> (co-supervisor). 2026–: in preparation, co-supervised with Dr. Jussi Karlgren.
Leadership	Responsible leader of a 6 MEUR Business Finland programme on trustworthy LLMs (2023–2024); Head of R&D at SiloGen, leading LLM development and co-PI of a 12 million GPU-hour LUMI allocation (2023–2024); work package lead in

the DeployAI consortium (2023); practice lead for Machine Learning and Computational Science at Accenture (2024–2025).

TEACHING MERITS

- 2024– Teacher, *Large Language Models and Generative AI for NLP*, Master’s level, University of Helsinki, every autumn since 2024, with Dr. Jussi Karlgren and Dr. Dmitry Kan. Designed the course syllabus; created and taught the lectures and coding labs for 4 of the 7 weeks (generative AI and LLM foundations, prompting, fine-tuning and the final project); assesses the final reports. Course materials openly available on GitHub (Helsinki-NLP).
- 2020 Co-instructor, *Natural Language Understanding and Representation Learning* (LDA-T3115), Master’s level, University of Helsinki, with Dr. Alessandro Raganato.
- 2019 Teaching assistant, *Machine Learning for Linguists* (KIK-LG210, BSc level) and *A Practical Introduction to Modern Neural Machine Translation* (MSc level), University of Helsinki. Lab monitor, Lisbon Machine Learning School (LxMLS 2019).

OTHER KEY ACADEMIC MERITS

Organising scientific events

- 2024– Co-organiser of ELOQUENT, a lab of shared tasks for evaluating generative language models at CLEF, since its first edition. CLEF 2024: organiser of the Topical Quiz task and co-organiser of the Robustness task. CLEF 2027: co-organiser of the proposed Fragments task (with Prof. Stergios Chatzikyriakidis) and of the Voight-Kampff and Robustness tasks.

Programme committees and reviewing

- Journal Natural Language Engineering (2022–).
- Conferences NAACL 2027 Demo Track; EMNLP 2024 Demo Track; COLM 2024; *SEM 2024; EMNLP 2021–2023; ACL-IJCNLP 2021; *SEM 2019–2021; NeurIPS 2021–2022; IJCAI 2020; ECAI 2019; NoDaLiDa 2019.
- Workshops UncertaiNLP at EMNLP 2025; BlackboxNLP at EMNLP 2024; UncertaiNLP at EACL 2024; BlackboxNLP and GenBench at EMNLP 2023; BlackboxNLP at EMNLP 2021; DL4NLP at NoDaLiDa 2019; RANLP 2019 Student Workshop.

Talks

Numerous talks at academic and industry events. A selection:

- 2022 Talks on multilingual, multimodal vector search at Berlin Buzzwords (with Dr. Dmitry Kan) and the ValleyML AI Expo.
- 2017–2022 Conference presentations at *SEM 2022 (Seattle), NoDaLiDa 2021 (Reykjavik) and NoDaLiDa 2019 (Turku); invited talk at CLASP, University of Gothenburg (2019); research seminar talks, University of Helsinki (2017–2019).
- 2018 Industry talks at Gartner Symposium, Barcelona, and AI Monday, Helsinki.

SCIENTIFIC AND SOCIETAL IMPACT

- Open science Code and data from the doctoral research released openly.
- Utilisation Bringing NLU and LLM research into practice: open Finnish language models at SiloGen, model customisation at Accenture, and AI inference at AMD Silo AI.

OTHER

- Languages Finnish (native), English (full professional), Swedish (limited working).
- Certificates Google Cloud ML and Data Engineer (2021); AWS Solutions Architect (2020).